Verifying Distributed Protocols with Proof Assistants: A Survey of Apporaches

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A clear and well-documented LaTeX document is presented as an article formatted for publication by ACM in a conference proceedings or journal publication. Based on the "acmart" document class, this article presents and explains many of the common variations, as well as many of the formatting elements an author may use in the preparation of the documentation of their work.

CCS Concepts: • **Security and privacy → Logic and verification**.

Additional Key Words and Phrases: Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 INTRODUCTION

ACM's consolidated article template, introduced in 2017, provides a consistent LaTeX style for use across ACM publications, and incorporates accessibility and metadata-extraction functionality necessary for future Digital Library endeavors. Numerous ACM and SIG-specific LaTeX templates have been examined, and their unique features incorporated into this single new template.

If you are new to publishing with ACM, this document is a valuable guide to the process of preparing your work for publication. If you have published with ACM before, this document provides insight and instruction into more recent changes to the article template.

The "`acmart`" document class can be used to prepare articles for any ACM publication — conference or journal, and for any stage of publication, from review to final "camera-ready" copy, to the author's own version, with *very* few changes to the source.

2 MOTIVATIONS

As noted in the introduction, the "`acmart`" document class can be used to prepare many different kinds of documentation — a double-anonymous initial submission of a full-length technical paper, a two-page SIGGRAPH Emerging Technologies abstract, a "camera-ready" journal article, a SIGCHI Extended Abstract, and more — all by selecting the appropriate *template style* and *template parameters*.

This document will explain the major features of the document class. For further information, the *LaTeX User's Guide* is available from https://www.acm.org/publications/proceedings-template.

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3 RESEARCH METHODS

3.1 Literature Search Strategy

To identify relevant papers, we used a systematic literature search strategy. We chose to query two databases, the ACM digital library and IEEE Xplore. These databases were accessed on April 17 2026.

To ensure the quality of of this review, we decided to follow the PRISMA framework to inform our search strategy. The search strategy can be summarized into multiple steps. First, we curated relevant keywords and constructed seach queries to compile relevant papers based on the guiding questions of the survey. This led to the following seach query:

("proof assistant" OR "theorem prover" OR "mechanized verification" OR "formal verification") AND ("distributed protocol" OR "distributed algorithm")

This query yeilded 213 results from the ACM Digital Library and 49 results from IEEE Xplore. [1]

3.2 Inclusion and Exclusion Criteria

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4 ONLINE RESOURCES

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REFERENCES

[1] Mauro Conti, Roberto Di Pietro, Luigi V. Mancini, and Alessandro Mei. 2009. (old) Distributed data source verification in wireless sensor networks. *Inf. Fusion* 10, 4 (2009), 342–353. https://doi.org/10.1016/j.inffus.2009.01.002

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